

Semantic Diff Prompting: Reducing Token Cost in Frame-by-Frame Video Understanding with Vision-Language Models

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Abstract—Vision-language models (VLMs) such as GPT-4o are increasingly used to describe video content by prompting the model independently on a sequence of extracted frames. This frame-by-frame approach is simple but wasteful: consecutive frames in a video are usually mostly identical, yet a naive prompting strategy re-describes static background and objects in every response, inflating token usage and API cost. This paper presents a small, controlled empirical study comparing a *baseline* prompting strategy (“Describe this frame.” applied independently to every frame) against a *semantic diff* prompting strategy, which instructs the model to describe only what changed relative to the immediately preceding frame. Holding the model fixed and varying only the prompt, we measure response-token cost using exact `tiktoken` (GPT-4 byte-pair-encoding) counts on a sample drawn from a 141-video, 8-category action-labeled dataset. We observe a 50–70% reduction in response tokens on non-first frames under the semantic diff strategy, without an obvious loss of dynamic information such as movement, new objects, or state changes, based on manual spot-checking. We report this result honestly as an empirical, sample-level observation rather than a population-level claim, and we discuss the threats to validity that limit generalization: single-model measurement, no downstream-task evaluation, no multi-frame temporal context, and no prompt-sensitivity sweep. We conclude that semantic diff prompting is a promising, low-effort prompt-engineering lever for reducing VLM video-understanding cost, and we outline the follow-up experiments needed to substantiate it as a general technique.

Index Terms—vision-language models, video understanding, prompt engineering, token efficiency, GPT-4o, tokenization

I. INTRODUCTION

Applying a vision-language model to video content typically requires converting the video into a sequence of still frames and querying the model once per frame, since most commercial VLM APIs (e.g., GPT-4o) accept image inputs rather than native video streams. This frame-by-frame querying pattern is attractive because it requires no specialized video-understanding architecture and works with any general-purpose multimodal chat model. However, it has an obvious inefficiency: real-world video is temporally redundant. A person walking across a static background produces frames that share almost all of their content — the background, lighting, and most objects do not change from frame to frame — yet a prompt such as “Describe this frame.” asks the model to regenerate a full scene description on every call, including the parts that have not changed. Because commercial VLM APIs charge per token, this redundancy directly inflates the dollar

cost of video understanding, and the cost scales linearly with the number of frames sampled from the video.

This motivates a simple question: if the model is instead asked to describe only *what changed* between consecutive frames, can token cost be reduced substantially while still capturing the information a downstream consumer would need (movement, new objects, state changes)? We refer to this alternative strategy as *semantic diff prompting*. The key design constraint is that this is a prompt-engineering intervention, not a model or architecture change: the model, sampling settings, and image encoding are held fixed, and only the natural-language instruction given to the model is varied.

This paper documents a small, honest empirical study of this question, built as a final project for a graduate generative-AI course. We implement both prompting strategies against the OpenAI GPT-4o family of models, measure token cost with an exact byte-pair-encoding tokenizer (`tiktoken`) rather than an approximate word or character count, and evaluate on a sample drawn from a 141-video action-recognition-style dataset spanning 8 categories. We report a 50–70% reduction in response-token count on non-first frames under the diff strategy. Consistent with the scope of a course project, we explicitly do not claim that this result generalizes across model families, video domains, or downstream tasks, and we describe the threats to validity in detail in Section VI.

II. RELATED WORK

Vision-language models. Large-scale vision-language models trained to jointly reason over images and text, such as GPT-4o, have made it practical to ask a single general-purpose model to describe, summarize, or answer questions about arbitrary visual content without task-specific fine-tuning. These models are most commonly exposed through image-input chat completion APIs, which is the interface this project targets.

Frame sampling for video understanding. Because most VLM APIs do not accept native video input, a common practical pattern for video understanding is to extract a subset of frames (e.g., every N th frame, or a fixed frame budget) and treat the video as a sequence of independent images. Frame-interval and frame-count sampling are the two knobs typically used to bound the cost of this approach, since cost scales with the number of frames processed rather than with

video duration. Dedicated video-native multimodal models (e.g., video-understanding models that accept a video clip as a single input) exist as an alternative to frame-sampling pipelines, but they solve a related, not identical, problem: they perform multimodal reasoning over a video directly, whereas the frame-sampling approach performs repeated single-image reasoning. This project is situated entirely within the frame-sampling paradigm and applies a token-cost lens to it rather than comparing against native video models.

Prompt engineering. A substantial body of practice has shown that the instruction given to a large language or vision-language model can materially change both the content and the length of its output, without any change to model weights. Prompt engineering interventions of this kind are attractive because they are cheap to test and deploy relative to model retraining or architecture changes. This paper treats the choice between a “describe everything” prompt and a “describe only the change” prompt as exactly this kind of intervention, and evaluates its effect on token cost using a fixed, held-constant model.

III. METHODOLOGY

A. Experimental design

The experiment is a controlled, within-subject comparison of two prompting strategies applied to the same frames from the same videos, using the same underlying model. Only the prompt text differs between the two conditions; the model, the image encoding, the maximum response token cap, and the sampling settings are held constant. This design choice — comparing prompts rather than models — was a deliberate architecture decision documented in the project’s decision record: the research question addressed is “does a diff prompt reduce token cost while preserving temporal information?”, which is a prompt-engineering question, not a model-selection question. Holding the model fixed isolates the effect of the prompt.

B. Baseline prompting

For the baseline condition, every extracted frame is submitted independently to the model with the fixed prompt “Describe this frame.” Each frame’s baseline response is generated with no reference to any other frame, so static content (background, fixed objects) is, in principle, re-described on every call.

C. Semantic diff prompting

For the diff condition, each pair of consecutive frames (previous frame, current frame) is submitted to the model with a fixed instruction: describe only what changed in the current frame relative to the previous frame, and do not repeat objects, background, or attributes that stayed the same. The instruction not to repeat unchanged content is the load-bearing part of the prompt; without it, the model tends to default back to a full re-description of the scene.

Because there is no preceding frame for the first frame in a sequence, the first frame’s diff response is not generated by



Fig. 1. An example frame-pair comparison generated by the pipeline, illustrating the visual basis for semantic diff prompting: the previous and current frames are shown alongside the regions of change that the diff prompt is intended to isolate, in contrast to the full-frame baseline description.



Fig. 2. A second representative frame-diff comparison from a different clip in the evaluated sample, further illustrating how consecutive frames share most of their visual content, motivating the token-cost hypothesis behind semantic diff prompting.

an API call at all. Instead it is replaced with a fixed synthetic string, “Initial frame. No previous frame to compare.” This design choice keeps the per-frame diff and baseline sequences the same length, which simplifies downstream comparison, output writing, and the accompanying automated tests, at the cost of the first frame’s diff “token count” being the tokenization of a fixed sentence rather than a genuine API response.

Fig. 1 and Fig. 2 show representative visualizations produced by the comparison pipeline, illustrating the frame-pair inputs and the corresponding semantic-diff output that the model is asked to generate against the baseline, full-frame description.

D. Token measurement

Token cost is measured using `tiktoken`, the byte-pair-encoding tokenizer matching GPT-4’s tokenizer, applied directly to each generated response string, with a GPT-2 tokenizer used only as a fallback in environments where `tiktoken` cannot be installed. This choice was made specifically because the entire comparison depends on token counts: using a tokenizer that does not match the inference model’s own tokenizer would invalidate the measurement. Reported token counts are computed over the response text only; the shared prompt text and the image token contribution of the multimodal payload are not counted in either condition, which is a fair comparison because both conditions share the same prompt structure and image encoding and therefore incur the same non-response token overhead. For each frame, the baseline token count and the diff token count are compared, and the percentage reduction is computed per frame and aggregated across frames.

IV. EXPERIMENTAL SETUP

The comparison pipeline accepts three kinds of input — a video file, a folder of images, or a single image — and extracts frames either via OpenCV video decoding (for video files, with configurable frame-interval and maximum-frame-count controls) or via sorted image loading (for image folders). The underlying dataset available for evaluation is a set of 141 action-labeled videos spanning 8 action categories, stored locally as a labeled video corpus; the reported results are drawn from a subset of this corpus rather than the full 141-video set, and no cross-validation or held-out test split was used. The model used for the comparisons is from the GPT-4o family, with `gpt-4o-mini` configured as the cost-bounded default and full `gpt-4o` available as an option; the model choice is held fixed across both prompting conditions within a given run. A fixed 3-second cooldown between consecutive API calls is used to stay within API rate limits, and a fixed response-length cap of 200 tokens bounds the maximum length of any single response in either condition. Frame sampling is controlled by two parameters — a maximum-frame-count cap and a frame-interval — that bound the number of frames, and therefore the number of API calls and the total cost, processed from any given video.

V. RESULTS

TABLE I
SUMMARY OF THE EMPIRICAL COMPARISON

Quantity	Value
Dataset size	141 action-labeled videos, 8 categories
Evaluation	Subset of the 141-video corpus
Prompting strategies compared	2 (baseline, semantic diff)
Tokenizer	<code>tiktoken</code> (GPT-4 BPE), GPT-2 fallback
Response token reduction	50–70% (non-first frames)
First-frame diff handling	Synthetic fixed string, no API call
Unit tests (deterministic pipeline)	18, no network calls

On non-first frames in the evaluated sample, semantic diff prompting produced a 50–70% reduction in response tokens relative to the baseline “describe every frame” prompting strategy. This reduction is consistent with the underlying hypothesis that static scene content dominates the token budget of a naive per-frame description, since the diff prompt’s only structural difference from the baseline is the instruction to omit unchanged content. Spot-checking of the diff outputs on the evaluated clips indicated that the diff responses continued to capture the dynamic content of interest — movement, newly appearing objects, and state changes — without hallucinating changes that had not occurred, though this check was qualitative rather than a scored evaluation. The first frame of each sequence is, by construction, not subject to this comparison in the diff condition, since its diff response is a fixed synthetic string rather than a model output; the 50–70% figure therefore applies specifically to non-first frames, where an actual diff response is generated and compared against an actual baseline response for the same frame.

As a supporting engineering signal rather than a result about the research question itself, the deterministic parts of the measurement pipeline (token counting, frame loading and sort order, API-key validation, base64 image encoding, and token arithmetic such as sums, averages, and percentage deltas) are covered by 18 unit tests that require no network access and no API key, and complete in well under one second; these tests do not validate the API request/response behavior itself, which would require an integration test against a live key.

VI. THREATS TO VALIDITY AND LIMITATIONS

We report this study’s limitations explicitly, following the honest-scope framing adopted in the project’s own documentation, which describes the work as “two prompting strategies compared on a small sample” rather than a generalizable benchmark result.

Sample size and statistical rigor. The underlying corpus contains 141 videos across 8 action categories, but the reported 50–70% figure is computed on a subset of this corpus, not the full set. No confidence intervals, no held-out test set, and no cross-validation were used. The 50–70% band is an observed empirical range on the evaluated sample, not a claim about the mean effect across the full population of videos.

Single-model comparison. Only one model family (GPT-4o, evaluated primarily via `gpt-4o-mini`) was used. Because the model was deliberately held fixed to isolate the prompt effect, this study makes no claim about whether the diff-prompting advantage transfers to other vision-language model families (e.g., Claude, Gemini, or open-weight vision-language models) without separate re-measurement on those models.

No downstream-task evaluation. This study measures token *cost*, not task *utility*. It is possible for the diff stream to be more token-efficient while being less useful as an input to a downstream consumer such as a video summarizer or a video question-answering agent; a complete evaluation would need

to measure both token cost and downstream task performance for each stream, which this study does not do.

No multi-frame temporal context. Each diff response is computed against only the immediately preceding frame; the pipeline does not use a sliding window or any temporal pooling across more than two frames. Consequently, longer-range temporal phenomena, such as an object or person re-entering the frame after being off-screen for several frames, are not correctly captured by a single-previous-frame diff.

Prompt sensitivity. Only one specific phrasing of the diff instruction was tested. Different phrasings are known, anecdotally, to produce materially different outputs from a vision-language model, and no systematic prompt-sensitivity sweep was performed to determine how much of the observed token reduction depends on this particular phrasing.

Non-determinism of the underlying model. GPT-4o was sampled with a default temperature that was not explicitly fixed, so repeated runs on the same input frames can produce different response text and therefore different token counts. The comparison in this study is only statistically meaningful across multiple repeated runs, which were not performed at scale due to API cost.

Not a substitute for native video models. This work is a frame-level summarization technique evaluated through a token-cost lens; it is not a claim that this frame-by-frame diff approach is equivalent to, or a replacement for, video-native multimodal models that accept an entire video clip as a single input. These represent different approaches to a related but distinct problem.

VII. CONCLUSION AND FUTURE WORK

This paper presented a small, controlled study comparing baseline frame-by-frame vision-language-model prompting against a semantic diff prompting strategy for video understanding, holding the underlying model fixed and varying only the natural-language prompt. Using exact byte-pair-encoding token counts, we observed a 50–70% reduction in response tokens on non-first frames when the model was instructed to describe only what changed between consecutive frames, evaluated on a subset of a 141-video, 8-category action-labeled dataset. We consider this an encouraging, low-effort prompt-engineering result rather than a validated general technique, given the study’s small sample, single-model scope, and lack of downstream-task evaluation.

Future work should directly address the limitations identified in Section VI: (1) evaluating the full 141-video corpus with repeated runs to obtain statistically meaningful confidence intervals on the token-reduction estimate; (2) repeating the comparison across additional vision-language model families to test whether the diff-prompting advantage generalizes beyond GPT-4o; (3) pairing both the baseline and diff output streams with a downstream consumer, such as a video summarizer or video question-answering agent, to measure whether token efficiency comes at a cost to task utility; (4) extending the diff computation to a multi-frame sliding window rather than a single-previous-frame comparison, to better

capture longer-range temporal coherence; and (5) performing a systematic sweep over diff-prompt phrasings to determine how sensitive the observed token reduction is to the specific wording used. Taken together, these extensions would move this study from a course-project-scale demonstration toward a reproducible benchmark for prompt-level cost reduction in frame-based video understanding.